CC105 WEEK 8 LABORATORY

Test Report – Stack-Based Expression Application

Programming Language: C++17

Testing Method: Manual testing through the program menu and expression evaluator.

Test Results

Test Input	Expected Output	Actual Output	Result
(12 + 3) * 2	Postfix: 12 3 + 2 *; Result: 30	Postfix: 12 3 + 2 *; Result: 30	PASS
8 + 2 * 5	Postfix: 8 2 5 * +; Result: 18	Postfix: 8 2 5 * +; Result: 18	PASS
20 / 5 / 2	Postfix: 20 5 / 2 /; Result: 2	Postfix: 20 5 / 2 /; Result: 2	PASS
10 - 3 - 2	Postfix: 10 3 - 2 -; Result: 5	Postfix: 10 3 - 2 -; Result: 5	PASS
{[2 + 3] * (4 + 1)}	Postfix: 2 3 + 4 1 + *; Result: 25	Postfix: 2 3 + 4 1 + *; Result: 25	PASS
7 / 2	Result: 3.5	Result: 3.5	PASS
0	Result: 0	Result: 0	PASS
5 / (3 - 3)	Division by zero error	Division by zero error	PASS
2 + * 3	Reject invalid operator sequence	Rejected: Invalid operator sequence or unary sign	PASS
2 3 + 4	Reject missing operator	Rejected: Missing operator	PASS
2(3 + 4)	Reject implicit multiplication	Rejected: Implicit multiplication rejected	PASS
()	Reject empty group	Rejected: Empty group or missing operand	PASS
-5 + 2	Reject unary sign	Rejected: Invalid operator sequence or unary sign	PASS
2.5 + 1	Reject decimal operand	Rejected: Decimal numbers are not supported	PASS
2 + a	Reject unsupported character	Rejected: Unsupported character	PASS
((2 + 3) * (4 + 5))	Result: 45	Result: 45	PASS
1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 + 9 + 10	Result: 55	Result: 55	PASS
5 + * 8	Reject invalid operator sequence	Rejected: Invalid operator sequence or unary sign	PASS

Stack Demonstration Test

The stack demonstration was run through menu option 3. The program reported an initially empty stack, safely rejected a pop operation on the empty stack, pushed 10, 20, and 30, displayed 30 as the top element, and popped the values in the order 30, 20, and 10. The final stack was empty with size 0. Result: PASS.

Menu Test

The menu was tested using options 1, 2, 3, and 0. Bracket checking, expression conversion and evaluation, stack demonstration, and program exit all worked as expected. Result: PASS.

Summary

Total test cases: 20. All recorded tests passed. The application correctly evaluated valid expressions and rejected the tested invalid expressions.

Note: The actual outputs in this report are based on the Programiz results provided during testing. Attach the corresponding screenshots in the screenshots PDF as supporting evidence.